



Tomás Sánchez Villaluenga

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PROFESSIONAL EXPERIENCE

LAST 4 ROLES
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Robotics Software Engineer

February 2026 - Present

Moffett Automation



- Lead **AI-driven process automation initiatives** across the company, reducing document review and update workload by **~90%** while significantly minimizing **data mismatches and critical client-facing errors**.
- Develop and integrate **Machine Vision features** to enhance system control, reliability, and operational performance.
- Design and implement **software solutions for Machine Vision pipelines and vision-based processes**.
- Develop and maintain **software for core robotics processes**, supporting the operation and continuous improvement of automated systems.

AI & Automation Consultant

May 2024 - July 2026

Santillana (Sanoma Group)



- Prepare proprietary original editorial content to create a new AI friendly dataset with previous content data to reduce friction with AI tools (**RAG, chunking, LLM-based tagging, knowledge graphs and data structuring**)
- Optimize **AI-assisted translation workflows**, reducing turnaround time from **~4 months to ~2 weeks** and costs by **~90%**, with potential savings of up to **98-99% vs. historical budgets**.
- Support the automation of **curriculum-adaptation workflows**, cutting processing time by **~50%** and costs by **~95%**, while bringing the process **in-house**.
- Advised and trained teams on **AI agents, Generative AI, automation and emerging AI technologies**, supporting their adoption across business and editorial workflows.

Robotics Engineer

October 2025 - January 2026

Dronomy



- Development and real-world implementation of **advanced perception and autonomy algorithms** for UAVs operating in **GNSS-denied environments**.
- Design and implementation of **collision avoidance** and **precision landing** algorithms to improve autonomous navigation, safety and robustness.

Critical Embedded Software Engineer

February 2023 - February 2024

Capgemini Engineering - Airbus Defense and Space



- Verification and validation of **safety-critical production software** for the **Airbus MRTT**, operating under **DAL-A** requirements and strict reliability standards.
- Contributed to **client-oriented software engineering delivery**, assessing software changes and system requirements to ensure **robustness, traceability and compliance**.

PRODUCTION AI & AGENTIC SYSTEMS

Agentic Document Inspector & Orchestrator – Sales & Engineering

[In production]

- Designed an **agentic document inspection and orchestration system** to identify contract-critical requirements and subtle specification changes across large engineering and commercial documentation.
- Automated **cross-document consistency and dependency management**, propagating changes from a single source document across related project documentation.
- Enabled AI-assisted **generation, validation and continuous updating** of engineering specifications, financial documentation, tender documents and technical schematics.

AI-Powered Real Estate Automation Platform

[In production]

- Developed an **end-to-end AI workflow** generating portal-ready property listings from structured data, adapting content to buyer profiles and target markets.
- Built a **multimodal computer vision pipeline** for room classification, alternate-view generation and virtual staging, including furniture removal/addition and style adaptation.
- Integrated **property pre-valuation** based on location, floor area and room characteristics.
- Developed a web interface orchestrating the full workflow from a customized property form to **one-click generation of listings and visual assets**.

PRODUCTION AI & AGENTIC SYSTEMS

HR & Workforce Management Platform

[In production]

- Developed a cross-platform **workforce management and time-tracking system** with Android/iOS apps and facial-detection-enabled kiosk check-in.
- Implemented intelligent monitoring for **shift inconsistencies, excessive working periods and missing rest intervals**.
- Built an **AI-driven Excel/report automation system** that reproduces company-specific templates and business logic using internal workforce data.
- Integrated a **digital fire register** using real-time attendance data to identify missing employees and support emergency workflows.

EDUCATION



BSc Electronics and Automation Engineering Universidad Politécnica de Madrid - Spain

September 2018 - June 2023

- Patent granted based on my Bachelor's Thesis (P202331053)
- Authored two papers on microelectronics and low-power electronics (Frontiers)
- Member of the Robotics Club and Sports Club



MSc Robotics and Automation Universidad Politécnica de Madrid - Spain

September 2023 - June 2025

- **Grade: 8.2 /10 (82%)**
- Relevant coursework: Computer Vision, Robot Guidance and Navigation, Optimal and Adaptive Control, Artificial Intelligence, Machine Learning, Deep Learning, Nonlinear Systems, Domotics, Advanced Automation.



Advanced Artificial Intelligence: Deep Unsupervised Learning and Computer Vision University of Oxford - United Kingdom (UK)

June 2025 - August 2025

- **Grade: A+**
- Deep Unsupervised Learning
- Generative AI
- Computer Vision
- Natural Language Processing

RESEARCH & PUBLICATIONS

- **Patent:** [Communicated system for detecting impacts and the status of an individual \(Ref.: P202331053\)](#)
- **Paper:** Geolocation and support electronic device for the humanitarian-military field (*Bachelor Thesis - UPM*)
- **Paper:** Wi-Fi/LoRa communication systems for fire, seismic-risk mitigation and health monitoring (*Collaboration with Materials Researching Institute of Madrid*)
- **Paper:** Event-Based Prediction Model for Multiple Objects and 3D Annotations in Logistics Scenario (*Master Thesis - Technische Universität Dortmund*)

LANGUAGES

Spanish - Native

English - C1

German - A2

TECHNICAL SKILLS

Software Engineering & Deployment

Python | C++ | C# | FastAPI | REST APIs | Pydantic | Git | Docker | Kubernetes | CI/CD | Microservices

Agentic AI & LLM Engineering

LangChain | LangGraph | CrewAI | Multi-Agent Systems | Agent Orchestration | Tool/Function Calling | State & Memory Management | ReAct | Human-in-the-Loop | MCP | Multi-LLM APIs (OpenAI, Anthropic, DeepSeek & Open-Source Models)

RAG & LLM Ops

RAG & Agentic RAG | Embeddings | Chunking | Vector Databases | Semantic Retrieval | Context Engineering | LLM Evaluation | Observability | Guardrails | Prompt Engineering | Cost & Latency Optimization

Machine Learning, Computer Vision & Robotics

PyTorch | OpenCV | YOLO | Deep Learning | Transformers & ViT | NeRF | ROS2 | SLAM | LiDAR | 3D & Event-Based Vision